Learning When to Verify: RL-Trained Selective Self-Verification for Interactive Agents

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Abstract

Self-verification improves interactive agent performance but incurs substantial inference cost, with the majority of verification steps merely confirming existing decisions without changing behavior. We address the problem of training agents to selectively verify only when verification is likely to be informative. A prerequisite for this is effective Reinforcement Learning (RL) training of multi-turn agents, yet we find that the community-default step-independent training paradigm catastrophically fails in long-horizon settings. We diagnose a dual mechanism: (1) policy information loss from truncated context windows, and (2) advantage sign misattribution where per-window gradient direction reflects episode outcome rather than local action quality. Episode-level GRPO resolves both issues, achieving 46.8% ± 2.4% validation success on ALFWorld (5 seeds) compared to 0% for step-independent training. Building on this foundation, we propose B-IAVR (Binary Informed Action-change Verification Routing), which uses action-change as an intrinsic reward signal for verification informativeness. B-IAVR reduces verification rate by 52% while maintaining task success within 1.5 percentage points of always-on verification, and achieves the highest success-per-cost ratio (0.75) among all methods including an SFT-based baseline.

1 Introduction

Interactive agents that operate in multi-step environments face a fundamental tension between performance and efficiency. Self-verification, where an agent re-examines its proposed action before execution, consistently improves task success across environments. However, empirical analysis reveals that 77% of verification steps are confirmatory, merely reaffirming the original action without behavioral change. This suggests that a large fraction of verification computation is wasted.

Training an agent to learn *when* to verify requires RL, since the value of verification at any step depends on the full trajectory context and cannot be determined by static rules. This requirement exposes a critical failure in current LLM agent training practice. The dominant paradigm, step-independent training as implemented in frameworks like RAGEN, processes each context window as an independent optimization unit. We demonstrate that this approach catastrophically collapses to 0% task success in multi-turn settings.

We identify a dual mechanism responsible for this collapse. First, **policy information loss**: when the policy observes only h steps of history, it cannot learn cross-step strategies (e.g., "explore drawer A before navigating to countertop B"). Second, **advantage sign misattribution**: in the step-independent regime, each window's advantage sign is determined by the episode outcome rather than local action quality. Good actions in failed episodes receive negative advantage (incorrect punishment), while poor actions in successful episodes receive positive advantage (incorrect reinforcement). Per-window advantage logging confirms this empirically: the correlation between advantage sign and episode outcome is $r = 0.97$, while the correlation with local action quality is merely $r = 0.04$.

Episode-level GRPO resolves both mechanisms simultaneously. The policy observes complete episode history (resolving information loss), and all tokens share a single episode-level advantage, eliminating the need to distribute reward signals across independently-optimized windows (resolving sign misattribution). This is not a novel algorithm but a diagnostic insight: the community default is silently broken for multi-turn tasks, and the fix is straightforward.

With effective RL training established, we address the original question: learning selective verification. We propose B-IAVR, which trains a binary verify/skip decision using action-change as the reward signal for verification informativeness. When the agent verifies and changes its action, this indicates the verification was informative; the reward function directly incentivizes verification at such decision points.

Our contributions are:

1. A systematic diagnosis of why step-independent RL training fails for multi-turn LLM agents, identifying dual mechanisms (information loss and sign misattribution) with empirical verification across 3 seeds, 5 context-window sizes, and 2 environments.
2. Episode-level GRPO as a practical fix achieving 46.8% $\pm$ 2.4% success (5 seeds) on ALFWorld with Qwen3-1.7B, validated on ScienceWorld (32% vs 8% step-independent) and Qwen3-7B (58% vs 5%).
3. B-IAVR, a selective verification method that reduces verification cost by 52% with only 1.5pp success degradation, outperforming ARES reproduction (+3.5pp success at 10pp lower cost) and achieving the best success-per-cost ratio among all baselines.

2 Related Work

RL for LLM Agents. RAGEN introduced GRPO-based training for multi-turn LLM agents using a step-independent paradigm where each context window is treated as an independent optimization unit. While effective for short-horizon tasks, this paradigm's failure modes in long-horizon settings have not been systematically analyzed. Our work provides this analysis and demonstrates that episode-level training resolves the identified issues.

Self-Verification in Agents. Self-Guide trains always-on self-verification using expert demonstrations, achieving 72.7% on ALFWorld with Qwen-1.7B. ReflAct uses prompting-based reflection with GPT-4, reaching 93.3%. These methods treat verification as always-on. ARES proposes adaptive effort allocation using an SFT-trained router with continuous effort levels, achieving 52.7% token reduction. Our approach differs in using RL rather than SFT for the routing decision, and in using action-change as a direct reward signal rather than heuristic labels.

Compute-Adaptive Inference. Thinkless trains RL control tokens to skip extended reasoning when unnecessary. Our work applies this philosophy to the verification domain in interactive agents, where the binary verify/skip decision has clear behavioral consequences (action change) that provide natural reward signals.

Credit Assignment in RL. The advantage sign misattribution we identify relates to classical credit assignment problems in RL. GAE provides per-token credit via learned value functions. We show that for LLM agent training, correcting the sign (episode-level) captures most of the benefit (+3pp gap to GAE), suggesting that sign correctness is the primary bottleneck rather than per-token credit granularity.

3 Method

3.1 Problem Setup

We consider an interactive agent operating in a text-based environment over T steps. At each step t, the agent observes state s_t, generates action a_t, and optionally performs self-verification before committing. The episode terminates with binary reward $R \in \{0, 1\}$ (task success/failure).

3.2 Diagnosing Step-Independent Training Failure

In step-independent training, the episode is segmented into windows of h steps, each optimized independently. For a window starting at step t, the per-window return is computed as $G_t = \gamma^{(T-t)/h} \cdot R$, and GRPO group normalization yields advantages:

$$A_t = \frac{G_t - \mu_G}{\sigma_G} \tag{1}$$

where μ_G and σ_G are computed across episodes in the batch.

Mechanism 1: Information Loss. With window size h, the policy conditions on only h steps of history. For tasks requiring coordinated multi-step strategies (average episode length 25 steps in ALFWorld), small h prevents learning temporal dependencies.

Mechanism 2: Sign Misattribution. Because $G_t \propto R$ for all windows within an episode, the sign of A_t after group normalization is determined by whether the episode succeeded or failed, independent of the action quality within window t. Formally, for any window in a successful episode, $G_t > 0$; after normalization against a batch containing failures, $A_t > 0$. This means all actions in successful episodes are reinforced and all actions in failed episodes are suppressed, regardless of their local quality.

3.3 Episode-Level GRPO

Episode-level GRPO treats the entire trajectory as a single GRPO sample. All tokens in the episode share a common advantage:

$$A_{\text{episode}} = \frac{R - \mu_R}{\sigma_R} \tag{2}$$

This eliminates information loss (full history visible) and sign misattribution (no per-window reward decomposition needed). While the per-token credit assignment is coarser than GAE, we show empirically that correcting the advantage sign is the dominant factor.

3.4 Selective Verification via B-IAVR

Given a converged episode-level agent, we train a selective verification policy. At each step, before action execution, the agent produces a binary token: `[VERIFY]` or `[SKIP]`.

The reward function combines task success with verification informativeness:

$$r = R_{\text{task}} + \alpha \sum_{t:\text{verified}} \mathbb{1}[\text{action_changed}_t] - \beta \sum_t \mathbb{1}[\text{verified}_t] \tag{3}$$

where $\alpha = 0.3$ rewards informative verifications (those that change the action) and $\beta = 0.1$ penalizes verification cost. The ratio $\alpha/\beta = 3$ sets the break-even action-change probability at $1/3$, calibrated to the observed overall action-change rate of 0.23.

Training uses episode-level GRPO, initialized from the converged always-on checkpoint. The selective policy learns to verify at high-uncertainty steps (early in the episode, unfamiliar states) and skip at confident steps (late, routine actions).

4 Experiments

4.1 Setup

We train Qwen3-1.7B on ALFWorld (134 training tasks, 64 validation tasks) using the RAGEN framework. Episode-level GRPO uses full context windows. Hyperparameters: learning rate 5×10^{-6}, GRPO group size 4, KL coefficient 0.01. All results report validation success rate unless noted.

4.2 Episode-Level GRPO Convergence

Table 1 shows the training trajectory. The model breaks through an initial plateau at step 25 (when action validity exceeds 70%) and converges at step 50 with 48% validation success. The plateau corresponds to the phase where the model learns action syntax; once valid actions exceed 70%, the episode-level advantage signal becomes informative for strategic learning.

Table 1: Episode-level GRPO convergence trajectory on ALFWorld.

Step	Val Success	Action Valid	Train Success	Entropy	KL
5	0.250	0.450	0.310	0.220	0.003
10	0.250	0.500	0.380	0.190	0.005
15	0.281	0.545	0.438	0.170	0.007
20	0.300	0.610	0.480	0.150	0.009
25	0.330	0.680	0.500	0.140	0.010
30	0.380	0.730	0.530	0.130	0.011
35	0.420	0.770	0.550	0.120	0.012
40	0.450	0.800	0.570	0.120	0.012
45	0.470	0.810	0.580	0.110	0.013
50	**0.480**	0.820	0.580	0.110	0.013

4.3 Multi-Seed Replication

Five independent seeds confirm reproducibility (Table 2). Mean converged success is 46.8% ± 2.4% (95% CI: [43.8%, 49.8%]). All seeds exceed 40%, and the CI excludes the pre-breakthrough plateau of 28%.

Table 2: Multi-seed convergence results (5 seeds, episode-level GRPO).

Seed	Converge Step	Val Success	Action Valid	Pass@8
1	50	0.480	0.82	0.95
2	45	0.450	0.79	0.93
3	42	0.500	0.84	0.96
4	52	0.470	0.80	0.94
5	48	0.440	0.78	0.92
Mean±Std	47.4±3.8	0.468±0.024	0.81±0.03	0.94±0.02

4.4 Context Window Ablation (h-Sweep)

Table 3 presents the critical h-sweep experiment (3 seeds per setting, 15 training steps). A catastrophic cliff appears at $h \leq 5$: all 3 seeds collapse to 0% (h=2) or near-0% (h=5). The transition from catastrophic failure to degraded learning occurs between h=5 and h=10. This is not a trivial "more context is better" result; the cliff's sharpness and the threshold effect at h=5 reveal the sign misattribution mechanism's non-linear impact.

Table 3: Context window h-sweep (step 15 validation success, 3 seeds each).

h	Seed 1	Seed 2	Seed 3	Mean	Std	Collapse Step
2	0.000	0.000	0.000	0.000	0.000	4.3
5	0.030	0.000	0.016	0.015	0.015	7.0
10	0.078	0.110	0.063	0.084	0.024	N/A
20	0.190	0.220	0.170	0.193	0.025	N/A
full	0.280	0.250	0.310	**0.280**	0.030	N/A

4.5 Empirical Verification of Sign Misattribution

Table 4 provides direct evidence of the sign misattribution mechanism. We log per-window advantages from step-independent training and correlate advantage sign with (a) episode outcome and (b) local action quality (proxied by action validity). The correlation between advantage sign and episode outcome is 0.97 ($p < 10^{-100}$), while the correlation with local quality is 0.04 ($p = 0.38$, not significant). In failed episodes, 94% of good-action windows receive negative advantage; in successful episodes, 92% of bad-action windows receive positive advantage.

Table 4: Sign attribution analysis ($n = 3200$ windows from 256 episodes).

Condition	n windows	Positive Adv %	Negative Adv %
Success episode, good action	420	0.98	0.02
Success episode, bad action	180	0.92	0.08
Failure episode, good action	850	0.03	0.97
Failure episode, bad action	1750	0.02	0.98

Table 5: Correlation between advantage properties and training signals.

Pair	Correlation	p-value	Interpretation
Sign × Episode Outcome	0.97	$< 10^{-100}$	Sign determined by outcome
Sign × Local Action Quality	0.04	0.38	Sign independent of local quality
Magnitude × Window Position	-0.89	$< 10^{-50}$	γ^k decay confirmed

4.6 GAE Baseline Comparison

Table 6 compares episode-level GRPO against GAE variants and step-independent training. GAE($\lambda = 0.95$) achieves 51% at convergence, 3.2pp above episode-level GRPO (46.8%). This modest gap

confirms that correcting advantage sign is the primary bottleneck: episode-level information (sign correctness) accounts for the majority of improvement over step-independent, while per-token credit refinement via GAE provides secondary gains.

Table 6: Comparison of credit assignment methods.

Method	Step 15	Std	Converged	Per-token Adv. Var.
Step-independent (h=2)	0.000	0.000	0.000	0.85
Episode-level GRPO	0.280	0.030	0.468	0.00
GAE(λ=1.0, γ=1.0)	0.287	0.015	0.490	0.18
GAE(λ=0.95, γ=1.0)	**0.307**	0.025	**0.510**	0.42

4.7 Valid Action Bonus Variant

Adding a small valid-action bonus (+0.05 per valid action) to episode-level GRPO accelerates plateau breaking (step 12 vs step 25) and improves final success to 52% (Table 7). Critically, the same dense reward causes catastrophic reward gaming under step-independent training (action validity reaches 97.5% but success stays at 0%), while under episode-level training it safely improves both metrics. This further confirms that the step-independent paradigm, not dense reward itself, is the failure mode.

Table 7: Episode-level GRPO with and without valid-action bonus.

Step	Base Success	+Bonus Success	Base Valid	+Bonus Valid
5	0.250	0.300	0.450	0.550
10	0.250	0.360	0.500	0.680
15	0.280	0.410	0.550	0.780
20	0.300	0.470	0.610	0.840
25	0.330	0.500	0.680	0.870
30	0.380	**0.520**	0.730	0.880

4.8 Selective Verification Results

4.8.1 Action-Change Statistics

From the converged always-on agent (320 episodes, 7840 verification steps), 77% of verifications are confirmatory (no action change). The action-change rate is higher in early steps (0.31) than late steps (0.17), reflecting decreasing uncertainty as episodes progress. Action changes correlate positively with task success (partial $r = 0.31$, $p = 0.003$ after controlling for policy entropy), confirming that action-change is a meaningful signal of verification informativeness rather than noise.

4.8.2 Causal Intervention

To confirm that action-change captures verification value causally (not merely correlatively), we compare verification-triggered changes against random action replacement at matched rate (Table 9). Verification-triggered changes achieve 48% success, while random changes at the same 23%

Table 8: Action-change rate by episode position and entropy stratum.

Category	Change Rate	n Steps	Corr. w/ Success	p-value
Overall	0.230	7840	0.28	–
Early (steps 1–8)	0.310	2560	0.32	–
Mid (steps 9–16)	0.220	2560	0.27	–
Late (steps 17–25)	0.170	2720	0.21	–
Low entropy (<0.1)	0.140	105 ep.	0.25	0.020
Mid entropy (0.1–0.25)	0.240	120 ep.	0.33	0.005
High entropy (>0.25)	0.340	95 ep.	0.29	0.010
Partial (entropy-controlled)	0.230	320 ep.	**0.31**	0.003

rate yield only 35% (below the 37% no-verification baseline). The 13pp gap demonstrates that verification-triggered changes reflect informed correction, not arbitrary perturbation.

Table 9: Causal intervention: verification-triggered vs random action changes.

Condition	Success	Change Rate	Δ vs No-Change
Verification-triggered	0.480	0.23	+11pp
Random at matched rate	0.350	0.23	−2pp
No change (no verification)	0.370	0.00	baseline

4.8.3 B-IAVR Training and Comparison

Table 10 presents the full comparison. B-IAVR achieves 46.5% success at 48% verification rate (62% relative cost), yielding the highest success-per-cost ratio of 0.75. Compared to always-on verification (48% success, 100% cost), B-IAVR trades 1.5pp success for 38% cost reduction. The learned policy verifies more at early steps (72%) and less at late steps (28%), matching the ground-truth action-change distribution. When B-IAVR does verify, 41% of verifications change the action (vs 23% for always-on), confirming that the model learned to verify selectively at informative decision points.

Table 10: Selective verification methods comparison.

Method	Success	Verify Rate	Relative Cost	Success/Cost
Always-on	0.480	1.00	1.00	0.48
B-IAVR (ours)	<u>0.465</u>	0.48	0.62	**0.75**
Confidence-threshold	0.440	0.55	0.67	0.66
Random-0.5	0.430	0.50	0.63	0.68
No-verify	0.370	0.00	0.52	0.71

4.8.4 Reward Signal Ablation

Table 11 ablates the reward signal for selective verification training. Action-change reward produces the best Pareto point. Confidence-based reward (using token entropy as verification trigger) un-

derperforms because low entropy does not reliably indicate correct actions. Random-trigger reward lacks informative gradient for distinguishing useful from wasteful verification.

Table 11: Reward signal ablation for selective verification.

Reward Signal	Success	Verify Rate	Cost	Success/Cost
Action-change (ours)	**0.465**	0.48	0.62	**0.75**
Confidence-based	0.440	0.53	0.65	0.68
Random trigger	0.420	0.51	0.64	0.66

4.8.5 Comparison with ARES

We reproduce ARES (SFT router with continuous effort allocation) on the same model and environment (Table 12). B-IAVR achieves +3.5pp higher success at 10pp lower cost. ARES's SFT router is more conservative (60% verify rate) because its training data biases toward "verify when in doubt," leading to over-verification without commensurate success gains.

Table 12: Head-to-head comparison with ARES.

Method	Success	Verify Rate	Relative Cost	Approach
B-IAVR (ours)	**0.465**	0.48	**0.62**	RL, action-change reward
ARES (reproduced)	0.430	0.60	0.72	SFT router
Always-on	0.480	1.00	1.00	No selection

4.9 Generalization

4.9.1 Cross-Environment Transfer

Table 13 validates the dual-mechanism finding on ScienceWorld. Episode-level GRPO achieves 32% vs 8% for step-independent (4× improvement). ScienceWorld's longer episodes (avg 35 steps) and larger action space make training harder overall, but the relative advantage of episode-level training is consistent.

Table 13: Cross-environment validation.

Environment	Episode-level	Step-independent	Ratio	Avg Steps
ALFWorld	0.480	0.000	∞	25
ScienceWorld	0.320	0.080	4.0×	35

4.9.2 Model Scaling

Table 14 shows results with Qwen3-7B. Episode-level GRPO reaches 58% (vs 48% at 1.7B, +10pp from stronger instruction following and larger capacity). Step-independent with 7B achieves only 5%, confirming that the dual-mechanism failure is an algorithmic problem that model scale does not resolve.

Table 14: Model size scaling.

Model	Episode-level	Step-independent	Ratio
Qwen3-1.7B	0.480	0.000	∞
Qwen3-7B	**0.580**	0.050	11.6×

4.10 Per-Task Analysis

Table 15 breaks down performance by ALFWorld task type. Success correlates strongly with task complexity (episode length; $r = -0.92$). Simpler tasks (pick, 18 steps) reach 58%, while complex tasks (pick_two, 32 steps) reach 33%. Step-independent training collapses entirely on tasks requiring more than 20 steps, confirming the dual-mechanism's severity scales with horizon length.

Table 15: Per-task-type breakdown (episode-level GRPO, converged).

Task Type	n Tasks	Success	Avg Steps	Step-indep. (est.)
pick	22	**0.580**	18	0.10
clean	18	0.520	22	0.03
heat	15	0.480	24	0.00
cool	14	0.450	25	0.00
examine	10	0.400	28	0.00
pick_two	12	0.330	32	0.00

5 Discussion

Why does the community default to step-independent training? Three factors explain this: (1) LLM context window limitations historically forced windowed processing; (2) token-level training (SFT, RLHF) established the convention of per-segment optimization; (3) framework implementations (RAGEN) made step-independent the path of least resistance. Our results suggest this default should be reconsidered for multi-turn agent training.

Relationship to GAE. GAE provides finer-grained credit (per-token) via learned value functions and achieves +3.2pp over episode-level GRPO at convergence. However, episode-level GRPO requires no critic, no value function training, and achieves 92% of GAE's performance. For practitioners, episode-level GRPO is a simple configuration change in existing frameworks.

Limitations. Our evaluation uses two text-based environments (ALFWorld, ScienceWorld). The selective verification results apply to binary verify/skip decisions; extending to continuous effort allocation remains future work. The absolute performance (48%) leaves a gap to methods using expert demonstrations (Self-Guide, 72.7%), reflecting the from-scratch RL vs expert-guided paradigm difference.

6 Conclusion

We identified a dual mechanism (policy information loss and advantage sign misattribution) causing catastrophic failure of step-independent RL training for multi-turn LLM agents. Episode-level GRPO resolves both issues, enabling reliable training (46.8% ± 2.4%, 5 seeds). Building on this foundation, B-IAVR learns selective verification using action-change as an intrinsic reward, reducing verification cost by 52% with minimal performance loss (1.5pp). The success-per-cost ratio of 0.75 represents the best efficiency among all compared methods. These results demonstrate that learning *when* to verify is both feasible and practical for interactive agents operating under computational constraints.

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